

This playbook outlines the protocols for our internal sandbox testing phase. It defines how test results are tabulated, how errors are logged and escalated, how role-specific checkmarks are assigned to job codes, how users interact with their agentic co-pilots, and how we assign those agents.

1. How Internal Testing Results Are Tabulated

Testing results are tracked and aggregated across three distinct layers, providing real-time auditability:

```
graph TD
    A[Test Execution] --> B[Layer 1: Central APM / Sentry Logs]
    A --> C[Layer 2: Local Firestore Test Ledger]
    A --> D[Layer 3: Intranet Dashboard Widgets]
    B --> B1[Raw Error Ingestion & X-Request-ID Tracking]
    C --> C1[Automated Script Runs & MAC Activation Matches]
    D --> D1[Operations Timeline & Full-Stack Audit Widget Updates]
```

1. **Centralized APM (Application Performance Monitoring) Logging:** Every transaction, API call, geocoding request, and claims state evaluation is tagged with a unique request header (X-Request-ID). If a test fails or encounters a timeout, Sentry routes a structured JSON log directly to our monitoring stack.
2. **Firestore Test Ledger (/test_runs):** A test coordinator script records automated test runs, capturing inputs, expected outputs, execution speeds, and pass/fail statuses.
3. **Visual Intranet Widgets (Consolidated View):**
 - [intranet_audit_widget.html]
(file:///C:/Users/greg_/OneDrive/Documents/Antigravity%20Save%20Folder/Chainmail-Intranet-Widgets/intranet_audit_widget.html): Displays the high-level health (0-100%) of all 13 full-stack architectural layers.

- [intranet_operations_timeline.html]
(file:///C:/Users/greg_/OneDrive/Documents/Antigravity%20Save%20Folder/Chainmail-Intranet-Widgets/intranet_operations_timeline.html): Displays a date-stamped log showing the specific ticket or test-case ID, description, and status (DONE, TO DO, QUEUED, PENDING_DVN).

2. What Happens When the System Hits an Error

To preserve capital and protect the network from incorrect state changes, we enforce strict fail-safe protocols:

1. Immediate Hard-Halt and State Exceptions:

- ***Geospatial Coordinate Drift:** If coordinates returned by geocoding API drift beyond the 0.0001-degree threshold from static imagery baselines, the system halts processing and raises a `GeospatialDriftException`.
- ***Confidence Intervals:** If multi-factor confidence calculations (evaluating local permits, municipality maps, and historical satellite tracks) drop below 95%, the automated underwriting engine halts, pausing active premium quotes.

1. The Adverse Non-Automation Rule (Human-in-the-Loop Handoff):

- Under Florida OIR and New York DFS Part 500 regulations, Agentic AI is strictly prohibited from executing autonomous adverse claims actions (denials or cancellations) without licensed human oversight.
- When an validation error or threshold breach occurs, the Agent cannot deny the file. Instead, the backend halts automated processing, marks the document status as `PENDING_ARBITRATION`, and pushes a alert panel containing raw JSON payloads directly to the human Integrity Officer's console.

1. **Backup Data Routing:** If a physical weather station (IoT gateway) goes offline or suffers telemetry packet loss, the Ingestion Agent registers a disconnect, switches to backup government weather feeds (NOAA/NWS) to check regional parametric bounds, and logs a recovery notice.

3. Job Code Testing Matrix

To maximize test coverage, coworkers will evaluate the platform features corresponding to their operational domains and job descriptions:

A. Executive Level (CEO / Strategic Oversight)

- Target Domain: Domain 4 (Treasury & Cryptographic Governance).
- Testing Focus:
 - Verify the 3-of-5 multi-signature threshold sequence for capital reserve tranches.
 - Verify Bitcoin treasury hedging triggers and check the USD-denominated digital reserve asset conversion ledgers.
 - Review high-level Intranet budget burn widgets for accuracy.

B. Operations & Product Management (COO / PMs)

- Target Domain: Domain 1 (Operational Lifecycle) and Domain 5 (IoT Ingestion).
- Testing Focus:
 - Simulate Shopify kit checkouts to verify the automated `/api/onboarding/provision` webhook post routes.
 - Test device MAC address pre-provisioning bindings in the Firestore sensors collections.
 - Trigger high-frequency simulated telemetry inputs to verify "verified node" transitions.

C. Legal, Risk & Compliance (CRO / Regulatory Officers)

- Target Domain: Domain 6 (Regulatory Compliance & Zero-Knowledge Privacy).
- Testing Focus:
 - Verify Zero-Knowledge proving outputs (π) to ensure no personally identifiable information (PII) or raw geocoded coordinates leak on public ledgers.
 - Test state compliance automated reports (e.g., FL OIR-B1-2222 metrics for timing constraints).
 - Audit the human-override adjuster panel to confirm the Adverse Non-Automation rule holds.

D. Underwriting & Actuarial (Lead Actuaries / Underwriters)

- Target Domain: Domain 2 (The Quant Brain & Parametric Pricing).
- Testing Focus:
 - Inject extreme coordinate drift inputs to confirm geocoding halt rules trigger.
 - Verify underwriting risk pricing results across different hurricane zones.
 - Stress-test confidence interval thresholds.

4. Agent Assignments & Interaction Mechanics

Every role within the Chainmail ecosystem is assigned an **Agentic Co-Pilot**. Users interact with their agents through unified intranet components.

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INTRANET WIDGET LAYOUT

[Operations Timeline] [Floating AI Knowledge Companion]

* OPS-257 (Done)

Connected:
Underwriting_Agent_01

INTRANET WIDGET LAYOUT

[Role Doctrine: Actuary]

User: "Check risk at 12665
Oaks"

Access: Read/Write Models

Agent: "Geocoding
coordinates..."

[Sidecar Action Panel]

[Send prompt] [Edit system
]

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How We Assign Agents

1. **Authentication Mapping (ABAC):** When a user logs in via Google OAuth, the system inspects their authenticated email against the `/users/{uid}` database registry.
2. **Role Provisioning:** Firestore assigns the user a corresponding role claiming tag.
3. **Agent Binding:** Based on the claiming tag, the user is paired with a specific backend Agent model:
 - Manager, AI Implementation $\rightarrow$ Antigravity (Core Agent)
 - Lead Actuary AI $\rightarrow$ Chief Underwriting Agent
 - Integrity Officer / adjuster $\rightarrow$ Claims Verification & Ingestion Agent

How Users Interact with Their Agents

- **Intranet Floating AI Chat Widget:** A persistent, gold-accented chat window is embedded in the Intranet workspace. The user chats directly with their assigned agentic helper, allowing them to query database logs, audit codebases, or run tasks.
- **Action Sidecars:** Inside the main dashboard, the Agent populates a sidecar pane with action cards (e.g., "Draft Claims Invoice Ready for Approval"). The human clicks "Approve/Reject" to complete the execution block.
- **Doctrine Access:** The [Operational Guidelines Widget] (file:///C:/Users/greg_/OneDrive/Documents/Antigravity%20Save%20Folder/Chainmail-Intranet-Widgets/intranet_roles_widget.html) links each role directly to their corresponding operational manuals and prompt structures, ensuring complete alignment.

5. Phased Sandbox Rollout Proposal

If you are not comfortable opening the system to all users immediately, we recommend a Phased Sandbox Testing Framework to isolate variables and maintain complete control:

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[Phase A: Greg Single-User] ---> [Phase B: Co-Founder Audit] ---> [Phase C: Department Sandbox] ---> [Phase D: Full Team Launch]

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- ****Phase A: Single-User Validation (Greg Only) – *Current****
- Only Greg's credentials have access rules in Firestore. No other users can log in or view databases. All live geocoding tests are run directly by Greg.
- **Phase B: Co-Founder Audit (2 Users)**
- Assign access specifically to your co-founder. Together, run sandbox transactions (simulated Shopify checkout, simulated claims trigger) to verify payments and multi-signature payouts.
- **Phase C: Phased Department Sandboxes (5 Users)**
- Enable access for the Lead Actuary and CRO. Restrict database access to a dedicated /sandbox` collection. Allow them to test underwritings and regulatory audits without write access to production database folders.
- **Phase D: Full Team Launch**
- Promote verified roles to standard production scopes, activate Google OAuth global logins, and monitor system integrations.

